

WASAGAMING, MB, Canada

WMO: 714440

Lat: **50.6550N**Lon: **99.9419W**Elev: **627**StdP: **94.01**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.8	-32.8	-39.6	0.1	-35.5	-36.4	0.1	-32.5	6.3	-12.3	5.5	-12.9	0.3	40	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	28.1	19.0	26.3	18.1	24.7	17.0	20.9	25.6	19.6	24.2	18.4	22.9	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	15.2	23.3	18.0	13.9	21.8	16.7	12.8	20.8	63.5	25.6	58.8	24.1	54.9	22.9	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.4	4.9	DB	-39.8	31.4	3.1	1.9	-41.9	32.8	-43.7	33.9	-45.5	35.0	-47.7	36.3	(4)
(5)			WB	-39.7	23.1	2.7	1.9	-41.7	24.5	-43.3	25.5	-44.8	26.6	-46.8	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-17.0	-15.0	-7.8	1.1	8.1	13.8	16.8	15.8	10.6	3.1	-6.4	-14.2	(6)
(7)		DBStd	13.42	8.27	7.88	8.03	6.22	4.74	3.87	3.21	3.39	4.46	4.81	6.68	7.66	(7)
(8)		HDD10.0	3971	837	700	553	269	93	9	0	1	44	221	492	751	(8)
(9)		HDD18.3	6435	1095	933	811	517	317	142	69	94	233	473	742	1009	(9)
(10)		CDD10.0	622	0	0	0	3	35	124	211	179	64	6	0	0	(10)
(11)		CDD18.3	46	0	0	0	0	1	7	21	14	2	0	0	0	(11)
(12)		CDH23.3	662	0	0	0	1	30	98	263	220	48	2	0	0	(12)
(13)		CDH26.7	126	0	0	0	0	5	16	47	52	6	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.7	1.8	2.1	2.4	2.5	2.2	2.0	2.1	2.3	2.1	1.6	(14)	
(15)	Precipitation	PrecAvg	552	24	17	37	30	73	96	71	66	49	45	24	24	(15)
(16)		PrecMax	690	35	38	133	61	159	164	163	168	83	137	89	51	(16)
(17)		PrecMin	424	9	0	9	9	21	37	29	13	5	12	3	5	(17)
(18)		PrecStd	72	8	9	30	17	36	38	35	39	21	28	22	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	4.5	12.4	21.1	26.8	28.7	30.1	30.9	27.3	21.7	11.9	4.0	(19)
(20)			MCWB	0.5	1.4	7.3	11.0	15.3	19.3	21.2	19.7	17.1	12.4	7.7	1.4	(20)
(21)		2%	DB	0.0	1.5	8.2	17.2	23.3	26.0	28.0	28.1	24.6	16.9	7.7	0.8	(21)
(22)			MCWB	-1.2	-0.5	4.0	8.5	12.8	17.4	19.6	19.1	16.1	10.0	3.9	-0.7	(22)
(23)		5%	DB	-2.2	-0.8	5.5	14.5	20.6	24.0	26.2	25.8	22.1	13.8	4.6	-1.8	(23)
(24)			MCWB	-2.9	-2.0	2.3	7.3	11.4	16.1	18.9	18.0	15.1	8.5	1.9	-2.8	(24)
(25)		10%	DB	-4.8	-3.1	2.7	11.6	17.9	22.0	24.5	23.8	19.4	11.0	2.2	-3.7	(25)
(26)			MCWB	-5.3	-3.9	0.6	5.7	10.2	15.3	17.8	17.0	13.6	7.0	0.5	-4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	1.9	7.1	11.6	16.7	21.9	23.4	22.4	19.7	13.1	7.5	1.5	(27)
(28)			MCDWB	2.1	3.6	10.9	18.8	22.6	27.0	27.2	27.2	23.8	18.8	11.9	3.5	(28)
(29)		2%	WB	-1.1	-0.3	4.3	9.5	14.3	19.1	21.1	20.3	17.6	11.1	4.4	-0.7	(29)
(30)			MCDWB	0.1	1.1	7.8	15.9	20.1	23.5	25.7	25.5	22.3	15.8	7.1	0.5	(30)
(31)		5%	WB	-3.0	-2.0	2.4	7.7	12.7	17.5	19.8	19.2	16.0	9.2	2.2	-2.7	(31)
(32)			MCDWB	-2.3	-0.8	5.1	13.3	18.7	21.8	24.3	24.0	20.6	12.8	4.5	-1.9	(32)
(33)		10%	WB	-5.2	-4.0	0.8	6.2	11.1	16.2	18.7	18.0	14.2	7.5	0.6	-4.3	(33)
(34)			MCDWB	-4.7	-3.1	2.6	11.5	16.4	20.4	23.1	22.4	18.5	10.7	2.0	-3.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	12.7	12.7	12.6	13.2	12.0	12.6	13.2	12.7	10.0	9.2	10.6	(35)
(36)			MCDBR	13.0	13.1	14.0	17.0	18.3	15.0	15.3	16.9	17.5	16.2	12.9	12.3	(36)
(37)			MCWBR	11.7	11.1	10.0	9.7	9.2	7.9	8.2	9.0	9.7	9.6	9.2	10.9	(37)
(38)		5% WB	MCDWB	12.6	12.2	12.8	15.6	15.0	12.5	13.2	13.8	14.7	14.3	12.1	11.8	(38)
(39)			MCWBR	11.5	10.4	9.4	9.3	8.5	7.3	8.0	8.2	9.0	8.8	8.9	10.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.252	0.298	0.328	0.341	0.354	0.369	0.366	0.333	0.282	0.253	0.227	(40)	
(41)		taud	2.280	2.305	2.289	2.396	2.419	2.406	2.379	2.383	2.462	2.574	2.479	2.373	(41)	
(42)		Ebn at Noon	794	900	907	917	915	902	880	865	862	863	805	787	(42)	
(43)		Edh at Noon	66	84	105	107	109	112	113	107	89	65	54	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.33	2.51	3.92	5.09	5.63	5.87	6.23	5.12	3.72	2.18	1.18	0.94	(44)	
(45)		RadStd	0.10	0.17	0.29	0.34	0.49	0.42	0.30	0.41	0.30	0.28	0.16	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page